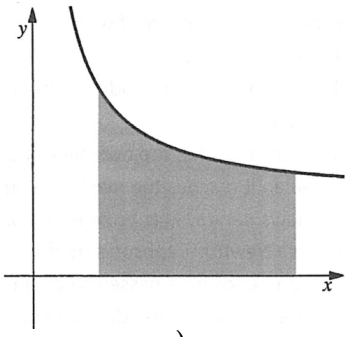
Integration: The Trapezium Rule

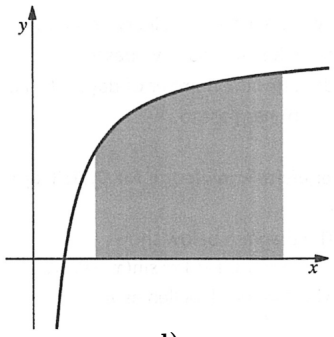
Questions 1–23

Question 1. The four shaded areas below are to be estimated using the trapezium rule. Which would produce an underestimate?

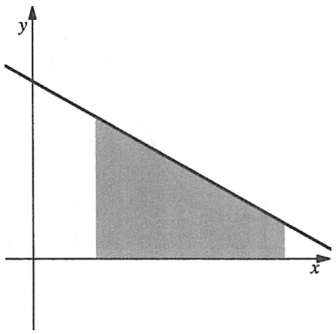
a)



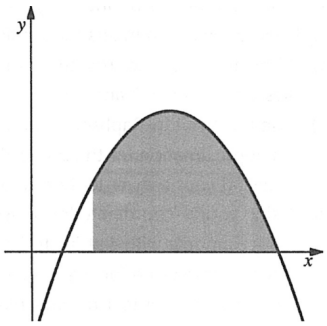
b)



c)



d)



Question 2. Amelia uses the trapezium rule to estimate an area from $x = 3$ to $x = 8$ using 9 ordinates. What is the horizontal width of each trapezium?

Question 3. Use the trapezium rule, with the stated number of strips, to estimate the following integrals. Give your answers in exact form.

a) $\int_0^3 x^2\,dx$, 3 strips

b) $\int_0^4 (x^2+1)\,dx$, 4 strips

c) $\int_0^2 x^3\,dx$, 4 strips

d) $\int_0^3 (x^3+2)\,dx$, 6 strips

Question 4. Use the trapezium rule, with the stated number of ordinates, to estimate the following integrals. Give your answers in exact form.

a) $\int_2^4 (2x^2+3x)\,dx$, 5 ordinates

b) $\int_1^2 (4-(x-1)^2)\,dx$, 5 ordinates

c) $\int_1^4 (20-x^2)\,dx$, 7 ordinates

d) $\int_{-3}^2 ((x+2)^2+3)\,dx$, 6 ordinates

Question 5. Use the trapezium rule, with the stated number of strips, to estimate the following integrals. Give your answers in exact form.

a) $\int_{-3}^{1}(x^3+2x^2+10)\,dx$, 4 strips

b) $\int_{-2}^{2}(5-(x-1)(x+2))\,dx$, 8 strips

c) $\int_{0}^{2}(3-\frac{1}{2}x^2)\,dx$, 5 strips

d) $\int_{-3}^{1}(2-\frac{1}{3}x^3)\,dx$, 5 strips

Question 6. Use the trapezium rule, with the stated number of ordinates, to estimate the following integrals. Give your answers in exact form.

a) $\int_{2}^{8}(8-x)(x-1)(x+2)\,dx$, 5 ordinates

b) $\int_{1}^{9}(3^x+2)\,dx$, 5 ordinates

c) $\int_{-7}^{-4}(\frac{2}{x+2}+2)\,dx$, 7 ordinates

d) $\int_{5}^{7}\frac{x+2}{x-4}\,dx$, 4 ordinates

Question 7. Use the trapezium rule, with four strips, to find an estimate for $\int_1^2 \frac{\ln x}{\sqrt{e^x+1}}\,dx$

Question 8. Given that $k > 0$, use the trapezium rule, with four strips, to show that:

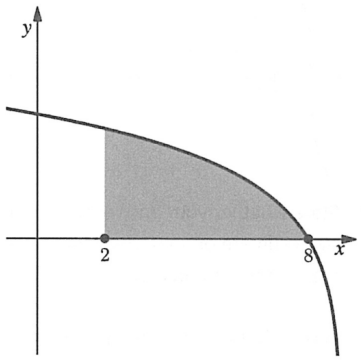
a) $\int_0^8 \frac{k+x^2}{2}\,dx = 4k + 88$

b) $\int_1^5 \frac{k+x}{x}\,dx = \frac{101k+240}{60}$

c) $\int_1^5 \ln(kx)\,dx = \ln(24\sqrt{5}k^4)$

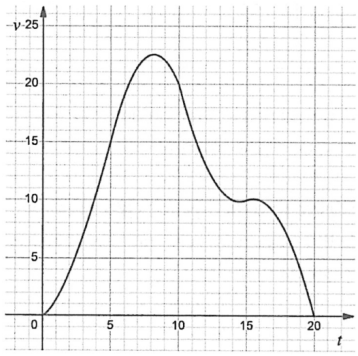
d) $\int_1^5 \ln(\frac{x}{k})\,dx = \ln 24 + \frac{1}{2}\ln 5 - 4\ln k$

Question 9. The curve shown below has the equation $y = \ln(9 - x)$.



Use three trapeziums to estimate the shaded region, giving your answer to three decimal places. Explain how you know this will give an underestimate.

Question 10. A particle is moving in a straight line with velocity v ms^{-1} at time t seconds as shown in the graph below. Use the trapezium rule with four strips to estimate the distance travelled by the particle in the first 20 seconds.



Question 11.

a) Use the trapezium rule, with four strips, to find an estimate for $\int_0^1 \frac{1}{\sqrt{4+x^2}}\,dx$

b) Explain how the trapezium rule might be used to give a better estimate to the integral given in a).

Question 12. Use the trapezium rule, with six ordinates, to find an estimate for $\int_0^1 \frac{x}{\sqrt{x^3+1}}\,dx$

Question 13. A particle is moving in a straight line and its speed is measured every 3 seconds. The results are given in the table below with the time in seconds and the speed in ms^{-1}. Use this information with the trapezium rule to estimate the distance travelled by the particle.

Time (s)	0	3	6	9	12	15	18
Speed (ms^{-1})	1	2.8	4.4	6.2	7.7	8.9	10.1

Question 14.

a) The table below shows the values of x and y for $y = \sqrt{\frac{2+x}{x}}$

x	2.5	3	3.5	4	4.5
y	1.342	1.291	1.254	1.225	1.202

The values of y are given to 4 significant figures. Use the trapezium rule with all the values of y in the table to estimate $\int_{2.5}^{4.5} \sqrt{\frac{2+x}{x}}\, dx$

b) Use your answer to deduce an estimate to $\int_{2.5}^{4.5} \sqrt{\frac{2+x}{2x}}\, dx$ giving your answer to 3 significant figures.

Question 15.

a) Use the trapezium rule, with four strips, to find an estimate for $\int_1^2 e^{3x^2}\,dx$

b) Use your answer to deduce an estimate to $\int_1^2 e^{3x^2+1}\,dx$

Question 16. The screenshot below shows part of a calculation to use the trapezium with five strips to estimate $\int_0^2 e^{8-x^3}\,dx$

	A	B	C	D
1	x	8-x^3	y	
2	0.0000	8.0000	2980.9580	1490.4790
3	0.4000	7.9360	2796.1535	2796.1535
4	0.8000	7.4880	1786.4756	1786.4756
5	1.2000	6.2720	529.5354	529.5354
6	1.6000	3.9040	49.6005	49.6005
7	2.0000	0.0000	1.0000	0.5000
8				6652.7439

Use the values in the spreadsheet to estimate the value of the integral to 3 significant figures.

Question 17.

a) The table below shows the values of x and y for $y = \log_5 \frac{x}{6}$

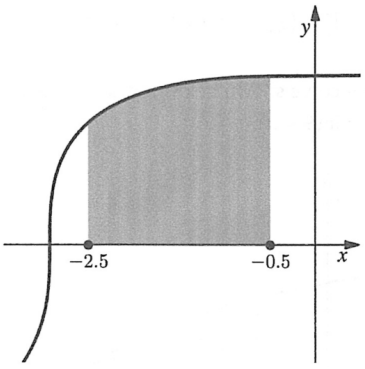
x	8	12	16	20	24
y	0.1787	0.4307	0.6094	0.7481	0.8614

The values of y are given to 4 decimal places. Use the trapezium rule with all the values of y in the table to estimate $\int_8^{24} \log_5 \frac{x}{6}\, dx$

b) Use your answer to deduce an estimate to $\int_8^{24} \log_5 \frac{x^2}{36}\, dx$

Question 18.

a) The diagram below shows the curve with equation $y = \sqrt[3]{x^3 + 25}$. Use the trapezium rule with five ordinates to estimate $\int_{-2.5}^{-0.5} \sqrt[3]{x^3 + 25}\, dx$



b) State whether your answer to a) is an overestimate or an underestimate, giving a reason for your answer.

Question 19. The screenshot below shows part of a calculation to use the trapezium with five strips to estimate $\int_0^6 \sqrt{e^{-x} - e^{-2x}}\, dx$

	A	B	C	D	E
1	x	exp(-x)	exp(-2x)	y	
2	0.0000	1.0000	1.0000	0.0000	0.0000
3	1.2000	0.3012	0.0907	0.4588	0.4588
4	2.4000	0.0907	0.0082	0.2872	0.2872
5	3.6000	0.0273	0.0007	0.1630	0.1630
6	4.8000	0.0082	0.0001	0.0903	0.0903
7	6.0000	0.0025	0.0000	0.0497	0.0249
8					1.0242

Use the values in the spreadsheet to estimate the value of the integral to 3 significant figures.

Question 20. Prerequisite knowledge for Q20: Radians Use the trapezium rule, with six ordinates, to find an estimate for $\int_0^\pi \sin^2 x\, dx$

Question 21. The table below shows the values of x and y for $y = \mathrm{f}(x)$

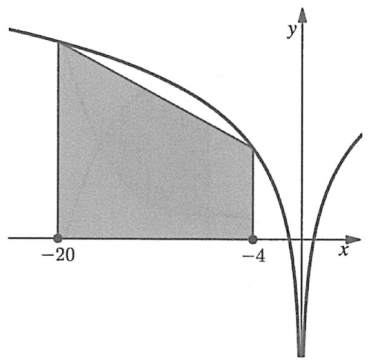
x	5	5.4	5.8	6.2	6.6
y	p	15	22	p	18

a) Use the trapezium rule with all the values of y in the table to estimate $\int_5^{6.6} \mathrm{f}(x)\,dx$ in terms of p

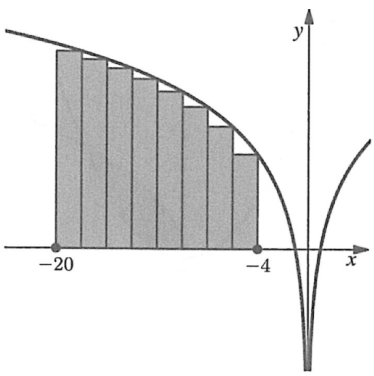
b) It is given that the estimate in part a) is 25. Find the value of p

Question 22. The diagram below shows the curve with equation $y = \ln(x^2)$

a) The area bounded by the curve, the x-axis, and the lines $x = -20$ and $x = -4$, is approximated by a single trapezium. Show that the area of the trapezium is $64\ln 2 + 16\ln 5$

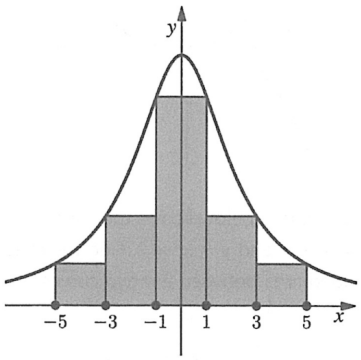


b) The area is now approximated using eight rectangles of equal width. Show that the total area of the rectangles is $60\ln 2 + 16\ln 3 + 4\ln 5 + 4\ln 7$

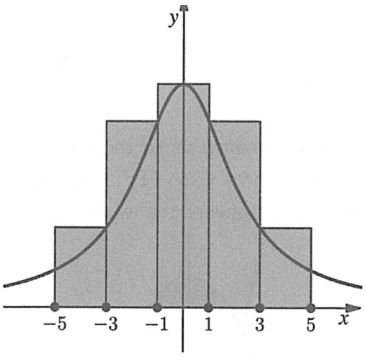


Question 23. Prerequisite knowledge: Integration by Substitution (for part d) Laura is trying to find an estimate for $\int_{-5}^{5} \frac{25}{5+x^2}\,dx$

a) She first uses five rectangles of equal width as shown in the diagram below. Show that this lower bound for the area is given by18.81 to 4 significant figures.



b) Laura then finds an upper bound for the value of the integral using five rectangles of equal width as shown in the diagram below. Determine the value of this upper bound to 4 significant figures.



Question 23 (continued). Prerequisite knowledge: Integration by Substitution (for part d) Laura is trying to find an estimate for $\int_{-5}^{5} \frac{25}{5+x^2}\,dx$

c) Laura then uses the trapezium rule with five strips to estimate the value of $\int_{-5}^{5} \frac{25}{5+x^2}\,dx$. Find her result to 4 significant figures.

d) Finally, Laura uses the substitution $x = \sqrt{5}\tan u$ to integrate and find an accurate value for the area. Find her result to 4 significant figures.